Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an Intelligent Multi-Joint Robotic Welding System for Mold Repair Using 3D Vision Sensors**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Artificial Intelligence (AI)*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Welding / Joining Machinery*	*Computer Vision / Image Processing*	*Object Recognition / Feature Extraction*
Concept of Proposed Technology	**Core Concept & Technical Overview** The technology proposed in this project aims to develop an **intelligent automated welding system for mold repair** that utilizes **3D vision sensors** to accurately recognize areas requiring repair welding and enables a **multi-joint robotic system** to automatically generate and execute optimal welding paths and process conditions. In this system, a skilled operator intuitively marks the areas on the mold surface that require repair welding. These markings are automatically detected using vision-based recognition techniques, and the detected weld regions are reconstructed into **3D geometric representations** that capture their shape and spatial location. Based on the reconstructed 3D information, robotic welding paths are automatically generated. During welding, **real-time weld position tracking and error**			

compensation are performed to ensure high-precision mold repair welding. After welding, **vision-based quality inspection techniques** are applied to automatically identify defects such as weld bead cracks and porosity, enabling systematic management of mold welding quality.

Through these capabilities, the proposed system simultaneously achieves **automation, high precision, and intelligence** in mold repair welding processes, reduces dependency on highly skilled welding labor, and improves welding quality consistency, productivity, and workplace safety.

Key Functions and Potential Applications

(1) Mold Repair Welding Position Recognition and Robotic Path Generation Using 3D Vision Sensors

• Automatic detection of operator-marked mold repair welding regions using cameras and 3D vision sensors

• Precise estimation of 3D position coordinates based on the extracted boundaries and geometric features of the weld region

• Automatic generation of optimal robotic welding paths considering mold geometry and collision avoidance

(2) Real-Time Weld Position Tracking and Error Compensation During Mold Welding

• Real-time tracking of the weld position using vision sensors during the welding process

• Real-time compensation for weld position deviations caused by thermal deformation and robot repeatability errors

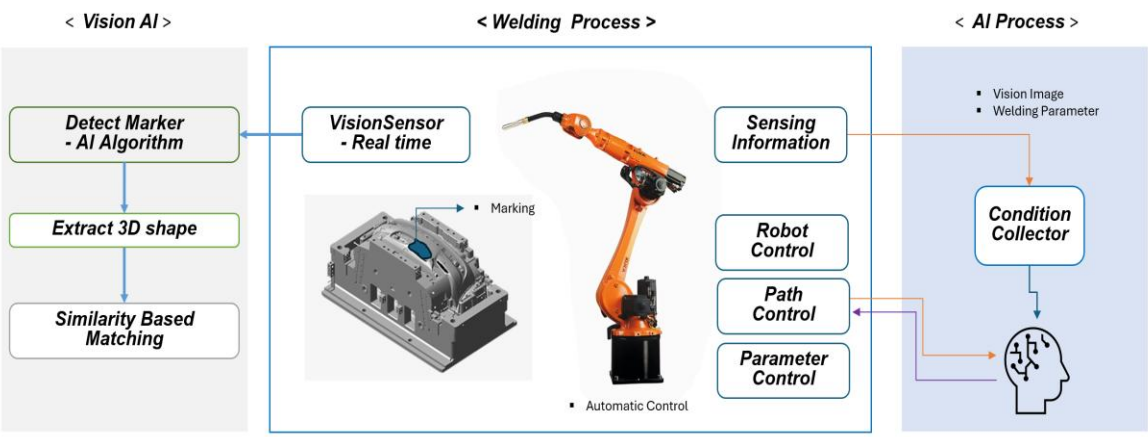
• Implementation of closed-loop (feedback) control integrated with robotic motion control for high-precision repair welding

(3) Post-Weld Quality Inspection for Weld Bead Cracks and Porosity

• Analysis of weld bead geometry using vision-based surface inspection techniques after welding

• Automatic detection of major defects in mold repair welding, including cracks, porosity, and undercut

• Digital recording of inspection results and implementation of a weld quality history management system



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Rationale for Joint R&D Project	**Critical Technical Barriers** Although domestic manufacturing companies possess extensive experience in mold and metal structure repair welding, along with strong capabilities in field deployment, quality evaluation, and process validation, significant technical limitations remain in the development of intelligent and automated welding systems. Advanced functions such as 3D vision-based position recognition, real-time weld seam tracking, error compensation, automatic robotic path generation, and vision-based quality inspection require sophisticated algorithmic design and high-level expertise in computer vision, robotics, and artificial intelligence. These capabilities cannot be sufficiently developed through field experience and conventional engineering approaches alone, creating barriers to achieving stable and high-precision automation in complex repair welding environments. **Why Global Collaboration?** Furthermore, the realization of a fully integrated smart welding system requires continuous validation through repeated on-site testing and iterative algorithm refinement based on real process data. However, domestic development environments are often limited in terms of specialized research manpower and access to advanced domain-specific methodologies in intelligent manufacturing, robotic automation, and AI-based inspection technologies. In particular, the development of high-precision perception, adaptive control, automated process planning, and intelligent quality assessment systems demands deep expertise in manufacturing AI and robotic intelligence, which cannot be sufficiently secured through conventional engineering approaches alone. In this context, collaboration with overseas research institutions possessing internationally recognized capabilities in these domains is essential. Such partnerships enable the transfer of advanced methodologies, reduce technological risks, and accelerate system optimization. Purdue University possesses world-class research capabilities in metal forming process physics modeling, digital twin–based manufacturing analysis, and physics-informed AI for manufacturing processes. In addition, Purdue operates specialized research centers such as the **Manufacturing and Materials Research Laboratory (MMRL)** and the **Center for AI and Robotic Excellence in Medicine (CARE)**, which provide advanced research infrastructure and interdisciplinary expertise in intelligent manufacturing, robotics, and AI-driven system development. These centers serve as critical platforms for integrating fundamental research with practical system implementation. As a manufacturing-focused engineering university located in Indiana, one of the major industrial hubs in the United States, Purdue maintains strong partnerships with medium- and large-scale manufacturing enterprises and actively utilizes large-scale equipment and production data for smart manufacturing and AI research. The university also operates testbed-based research environments and benefits from its strategic location within a major transportation network. This enables direct observation of technological trends and market demands in the U.S. manufacturing sector. Consequently, collaboration with Purdue University provides a unique opportunity to establish a comprehensive development framework that links industrial needs with core technology

	development, field validation, system refinement, and commercialization, thereby enhancing technological maturity, field applicability, and global competitiveness.
Key Words	1. **User-Marked 3D Weld Recognition** 2. **Deep 3D Perception** 3. **AI-driven Welding Parameter Optimization** 4. **Adaptive Robotic Welding** 5. **Vision-AI Seam Reconstruction**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)

R&D Objectives

R&D Objectives

The objective of this project is to develop an intelligent automated mold repair welding system that integrates 3D vision-based position recognition, automatic robotic path generation, real-time weld tracking with adaptive error compensation, and vision-based post-weld quality inspection to ensure high-precision, reliable, and repeatable mold repair welding operation.

Detailed Objectives

This project aims to develop an integrated smart mold repair welding system by combining advanced sensing, robotics, and artificial intelligence technologies. First, the system will accurately detect operator-designated mold repair areas using cameras and 3D vision sensors and extract geometric boundaries and surface features to estimate precise three-dimensional coordinates of weld regions. Based on this spatial information, optimal welding paths will be automatically generated for multi-joint robotic systems while considering mold geometry and potential interference with surrounding structures, thereby ensuring stable and repeatable welding performance.

During the welding process, the system will implement vision-based real-time weld position tracking and closed-loop adaptive robotic control to compensate for positional deviations caused by thermal deformation and reduced robot repeatability. This adaptive control framework will enable high-precision welding under varying operating conditions. After welding, vision-based surface inspection techniques will be applied to analyze weld bead geometry and surface conditions and to automatically detect major defects such as cracks, porosity, and undercut. All inspection and process data will be digitally recorded and managed to support weld quality traceability, continuous process optimization, and long-term reliability improvement of mold repair welding operations.

R&D Target Specifications

Item	**Target Performance**
User Marking Recognition Accuracy	≥ 95%
Weld Seam Position Detection Error	≤ ±0.5 mm
Robotic Welding Path Generation Success Rate	≥ 98%

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Welding Defect Rate	≤ 5%
Average Task Time Reduction in Field Application	≥ 30%

R&D Plan

Phase 1: Core Technology Development for 3D Vision–Based Mold Repair Welding Position Recognition and Robotic Path Generation (Year 1)

• Definition of operator marking methods and establishment of a data collection framework for mold repair welding markings
• Development of vision-AI–based algorithms for operator marking recognition and automatic detection of weld locations using cameras and 3D vision sensors
• Development of precise 3D position coordinate estimation techniques based on extracted boundaries and geometric features of detected weld regions
• Development of automatic robotic welding path generation algorithms for multi-joint robots, considering mold geometry and interference with surrounding structures
• Verification of collision avoidance and robot reachability of generated welding paths in a robotic simulation environment

Phase 2: Integration of Real-Time Weld Position Tracking and Adaptive Robotic Control and Prototype Development (Year 2)

• Development of multi-joint robot control interfaces and real-time control software
• Development of vision sensor–based real-time weld position tracking algorithms
• Implementation of real-time weld position error compensation logic to address thermal deformation and robot repeatability errors
• Integration of a closed-loop (feedback) control system linking vision recognition, robotic path execution, and real-time error compensation
• Fabrication of an intelligent automated mold welding system prototype
• Validation of welding accuracy, repeatability, and process stability in a laboratory environment

Phase 3: Post-Weld Quality Inspection and Industrial Field Demonstration and System Enhancement (Year 3)

• Deployment of the system in actual mold repair welding production environment and establishment of field demonstration infrastructure
• Repeated field demonstrations on molds with various geometries to validate position recognition, tracking, and error compensation performance
• Implementation of vision-based surface inspection techniques for weld bead geometry analysis after welding
• Advancement of automatic defect detection algorithms for major mold repair welding defects, including cracks, porosity, and undercut

	• Establishment of digital recording and weld quality history management systems • Finalization of system performance improvements and commercialization-ready system design based on field demonstration results

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Purdue University research team will be responsible for developing core algorithms and control technologies for the intelligent welding system: • Developing algorithms for automatic weld region detection, boundary and geometry analysis, and precise three-dimensional position estimation using cameras and 3D vision sensors • Developing automatic robotic welding path generation algorithms for multi-joint robots that consider mold geometry and potential structural interference • Developing real-time weld position tracking and closed-loop adaptive control technologies to compensate for thermal deformation and robot repeatability errors • Developing automatic defect detection algorithms based on weld bead geometry analysis and supporting field demonstration and on-site deployment activities
Expected R&R of Korean SME Partner	Korean SMEs, as the lead institution of this joint research and development project, will be responsible for overall project management and field-oriented demonstration of the developed technologies: • Defining technical and operational requirements for actual mold repair welding processes and providing diverse mold and metal component samples for demonstration data collection • Supplying production-site testbeds and equipment environments to support deployment of multi-joint robotic automatic welding systems and prototype installation • Conducting repeated field demonstration trials to evaluate welding accuracy, operational efficiency, process stability, and weld quality, and providing structured feedback for system improvement • Leading the development of commercialization-ready models and establishing business and market deployment strategies for practical industrial application

5. Expected Outcomes/Impact

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Expected Outcomes and Impact	Through this joint research and development project, the core enabling technologies of an intelligent automated mold repair welding system will be established by integrating 3D vision sensor–based weld location recognition, automatic robotic path generation, real-time weld position tracking and error compensation during welding, and post-weld vision-based quality inspection. By developing and demonstrating a prototype system suitable for industrial environments, the project will advance the technology from TRL 3 (proof of concept) to TRL 6 (industrial demonstration). This is expected to overcome the limitations of conventional manual mold repair welding processes, including operational variability, inconsistent quality, and heavy reliance on highly skilled labor, while enabling stable implementation of high-precision and high-quality welding processes. From a technical perspective, this project is significant in that it secures fundamental technologies for intelligent welding automation through the integrated application of operator-marked weld location recognition, vision-AI–based 3D weld detection and position estimation, automatic robotic path generation considering mold geometry and interference, adaptive robotic control based on real-time weld position tracking and error compensation, and post-weld automatic inspection of weld bead cracks and porosity. These technologies constitute a core technical foundation that can be extended beyond mold manufacturing to a wide range of applications, including complex-shaped metal components, high-mix low-volume production processes, and precision-critical industrial equipment and structural manufacturing. In particular, by digitizing and automating mold repair welding processes that traditionally depend on the experience of skilled workers, the proposed system offers a highly adaptable and field-deployable automated welding solution with differentiated technological competitiveness. From a commercialization perspective, the automated welding system demonstrated in the actual mold repair welding production environment of the lead institution, Korean SME, can be further developed into a commercialization-ready model and deployed incrementally across various manufacturing sites, starting with mold manufacturing companies and expanding to metal structure and industrial equipment component manufacturers. Through this, the project is expected to contribute to improved productivity, enhanced welding quality consistency, reduced labor costs, and improved workplace safety in domestic manufacturing industries. Furthermore, it is anticipated to generate significant industrial impact by creating new markets for 3D vision–based intelligent mold welding equipment and robotic welding systems.

6. References

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